ORIGINAL ARTICLE

Camouflage, memory, and truth-sensitive word of mouth in fake-news dissemination: a seeded agent-based simulation study

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ARTICLE HISTORY
Compiled August 14, 2026

ABSTRACT
Fake-news sources can imitate established media without changing how often they publish false content. We study this mechanism with FAKENEWS-ABM, an endorsement-based agent model in which users learn about sources, select one source per period, and exchange word-of-mouth (WOM) recommendations. Across 2,903 runs, we separate source popularity from actual false reposting and examine copy scope, intervention timing, memory, truth-sensitive WOM, contacts, and reach. The baseline reproduces configured false-publication probabilities. Penalising false recommendations while rewarding true ones reduces late-run false reposts by 5.26 percentage points relative to discovery-only WOM. Full attribute copying produces the largest target-selection gain (8.54 points) but reduces false reposting because credibility also controls objective false-publication probability. Early copying of all non-credibility attributes instead preserves the target's false-news propensity. A 30-seed confirmation finds a 2.89-point increase in false reposts under memory 25, false-news penalisation, and no reward for true news. Effects weaken with delayed intervention or short memory. Common seeds reproduce condition rankings and usually narrow uncertainty. The results show why popularity is an inadequate misinformation proxy and identify memory, timing, and credibility coupling as central modelling considerations.

KEYWORDS
agent-based simulation; fake news; misinformation; word of mouth; endorsements; memory; scenario analysis

1. Introduction

Online misinformation is simultaneously a content problem, a social-influence problem, and a measurement problem. False claims can reach large audiences, yet exposure and sharing are not evenly distributed across people or sources (Lazer et al., 2018; Vosoughi et al., 2018; Grinberg et al., 2019; Guess et al., 2019). A source may become more visible because it uses emotionally engaging language, familiar visual conventions, or socially reinforced recommendations. However, a source's popularity does not by itself indicate how much false content it contributes. This distinction becomes especially

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important when a low-credibility source adopts the outward characteristics of an established news organisation. The source may gain attention while retaining its original editorial propensity to publish false material, or it may imitate both presentation and credibility and thereby change the very process that generates false content.

These mechanisms are difficult to isolate with observational platform data. Algorithms, user preferences, social ties, source reach, and content production evolve together. Ethical and practical constraints also prevent researchers from deliberately increasing the attractiveness of a misinformation source in a live platform. Agent-based simulation offers a complementary laboratory in which the mechanisms are explicit, interventions are reproducible, and alternative configurations can be compared while holding other elements constant (Macal and North, 2010; Railsback and Grimm, 2019). Such models do not replace empirical research. Their value lies in making assumptions operational, exposing interactions, and generating conditional claims that can later be calibrated and tested.

This study develops that laboratory using FAKENEWS-ABM, an agent-based model of source evaluation and reposting on social-network sites. The reasoning mechanism is based on endorsement theory (Cohen, 1985). Users accumulate positive or negative evidence about multiple dimensions of a source, combine that evidence with individual attribute weights, and select a source probabilistically. Word of mouth (WOM) transmits an additional endorsement from a user's contacts. The model extends an earlier endorsement-based line of simulation research on electronic marketplaces, where choices, recognition, and WOM jointly produce market shares (Terán et al., 2022, 2024). In the present application, a source selection represents a repost, and every source independently publishes a true or false item in each period. This makes it possible to distinguish the share of selections received by a target source from the share of selections attached to an item that was actually false.

The research was motivated by a scenario in which a fake-news source copies attributes from a traditional source. Initial experiments treated copying as an undifferentiated operation. Inspection of the model then revealed a consequential dependency: the source-credibility attribute affects both user evaluation and the probability that a source publishes false news. Copying all attributes therefore changes two mechanisms at once. It makes the target resemble the donor in presentation and also makes the target objectively less likely to publish a false item. That scenario remains useful as a mechanistic diagnostic, but it is not a clean representation of strategic camouflage. A more informative treatment copies all attributes except credibility, thereby changing competitive presentation while retaining the target's original false-news propensity.

Memory and WOM add temporal and social dependencies to this distinction. A copied attribute cannot influence a user indefinitely when endorsements expire quickly. An intervention at period 100 may have less opportunity to shape later evidence than an intervention at period 1. WOM can also act in several ways. It may simply reveal an otherwise unknown source, it may penalise a recommender's source after a false publication, or it may reward that source after a true publication. These policies are not interchangeable. A zero outcome is implemented as no outcome endorsement, not as a negative value. Consequently, the model can represent four substantively different rules: penalise or ignore false recommendations, and reward or ignore true recommendations.

We consolidate three experiment phases comprising 2,903 simulation runs. An exploratory phase studies copy scope, memory, timing, network conditions, and common random seeds. A separate baseline phase tests configured false-publication rates and compares no WOM, discovery-only WOM, and truth-sensitive WOM. Finally, strategy, confirmation, and robustness suites examine copied-attribute scope, outcome-specific

WOM policies, contact degree, and source reach. The phases are not pooled as if they were identical samples; they are used as complementary levels of evidence. Broad factorial results are treated as exploratory, whereas a smaller set of policy contrasts is repeated with 30 common seeds.

The study addresses four research questions:

(1) Does the model's baseline reproduce configured false-publication rates, and does truth-sensitive WOM reduce false reposting relative to discovery-only WOM?
(2) How do copied-attribute scope, memory horizon, and activation time affect target-source selection and actual false reposting?
(3) How does the treatment of false and true WOM outcomes alter the effect of non-credibility camouflage?
(4) Are the direction and ordering of the findings robust to contact degree, source reach, and common-seed replication?

The contribution is threefold. First, we provide an operational separation between popularity and realised misinformation in an endorsement-based simulation. Second, we show that a modelling shortcut—using credibility both as a perceived attribute and as an objective content-generation parameter—can reverse the apparent implication of an intervention. Third, we provide a reproducible seeded experiment harness and a validation strategy that distinguishes baseline behaviour, exploratory ranking, and confirmatory contrasts. The resulting claims are about the implemented model, not direct causal estimates for a real platform. Nevertheless, they identify mechanisms and measurement choices that matter for future empirical calibration and for simulation studies of misinformation.

The rest of the paper is organised as follows. Section 2 reviews misinformation diffusion, social influence, and endorsement-based reasoning. Section 3 describes the agents, sources, decision process, WOM policy, memory, and scenarios. Section 4 presents the experimental design and validation procedure. Section 5 reports baseline, strategy, policy, and robustness results. Section 6 discusses their interpretation and limitations, and Section 7 concludes.

2. Theoretical background

2.1. *Misinformation as exposure, choice, and diffusion*

Research on online misinformation distinguishes production, exposure, belief, and sharing. These events are related but not identical. The existence of false content does not imply that every user is exposed to it, and exposure does not imply belief or reposting. Empirical studies show that false-news consumption and sharing can be concentrated among relatively small groups (Grinberg et al., 2019; Guess et al., 2019). At the same time, false stories can travel farther and faster than truthful stories under particular platform and content conditions (Vosoughi et al., 2018). This combination motivates metrics that retain both the source of a decision and the truth state of the item selected.

Source credibility offers a useful but incomplete heuristic. Crowdsourced source-quality judgments can help identify unreliable outlets (Pennycook and Rand, 2019), and accuracy prompts can redirect attention toward truthfulness (Pennycook et al., 2020; Epstein et al., 2021). Yet source labels and reputations operate alongside content-level features. Emotion, novelty, simplicity, proximity, sensationalism, multimedia, and other engagement cues may shape selection before a user knows whether the item is true. Moreover, a source can alter its style without altering its underlying editorial behaviour. A model that collapses perceived credibility and actual truth generation into one variable risks obscuring this difference.

The wider information environment also matters. Algorithmic and individual choices jointly structure exposure (Bakshy et al., 2015). Homogeneous communities can reinforce selective narratives (Del Vicario et al., 2016), while social influence can bias evaluations even when the initial social signal is arbitrary (Muchnik et al., 2013). Network experiments further show that diffusion depends on reinforcement structure and the complexity of the behaviour being transmitted (Centola, 2010). These findings do not determine a unique simulation mechanism, but they establish that content attributes, social transmission, and network opportunity should be represented separately enough to support diagnosis.

2.2. *Endorsements as bounded evidence*

Endorsement theory represents reasoning under uncertainty through labelled items of evidence whose interpretation depends on their source, recency, and relevance (Cohen, 1985). Instead of assigning a single immutable utility to an alternative, an agent accumulates endorsements and evaluates them in context. This is attractive for social simulation because users can hold heterogeneous weights, observe incomplete information, and update source evaluations over time.

The prior e-commerce model by Terán et al. (2022) uses endorsements to represent marketplace choice and WOM. Buyers evaluate marketplace attributes, retain evidence, and communicate choices. Terán et al. (2024) extends that work through scenario analysis and repeated Monte Carlo experiments, emphasising how recognition, attributes, and WOM shape market share. FAKENEWS-ABM preserves the reasoning architecture while changing the domain interpretation. News sources replace marketplaces; users replace buyers; a selected source denotes a repost; and source-level truth states make it possible to calculate realised false reposts.

This transfer creates an important conceptual caution. Market share is a suitable outcome when the primary phenomenon is marketplace dominance. In misinformation research, source share is only one layer. A source can dominate while publishing mostly true content, or remain relatively small while producing a high proportion of false items. Accordingly, the present study treats target-source selection as an intermediate outcome and actual fake-repost share as the primary dissemination outcome.

2.3. *WOM, truth feedback, and memory*

WOM is interpersonal transmission about an alternative. In the model it performs two logically distinct functions. First, a recommendation can make an unknown source available for later evaluation. Second, after the recommended publication's truth state is known, the receiver can add a signed endorsement to that source. The first function is discovery; the second is truth-sensitive social learning. Keeping them separate allows a baseline test: if discovery-only WOM and no WOM are identical, any difference created by truth-sensitive WOM can be attributed within the model to the signed outcome rule rather than to source discovery alone.

Signed feedback is configurable. A false publication can be penalised (-1), ignored (0), or rewarded ($+1$); a true publication can be treated in the same way, although the current experiments focus on ignoring or rewarding true outcomes. Ignoring means that no outcome endorsement is created. It does not store a zero-valued endorsement that might later be recoded as negative. This distinction is necessary because a binary conversion of zero to “not positive” would silently turn neutrality into punishment.

Memory determines how long accumulated endorsements remain active. Short memory creates rapid turnover: copied attributes can influence only recent observations, and an intervention needs repeated reinforcement to persist. Long or infinite memory allows early evidence to remain in the decision set. The resulting effects need not be monotonic. Long memory can preserve treatment-generated endorsements, but it can also preserve pre-intervention evidence or stabilise a control. Therefore memory must be analysed jointly with activation time and WOM policy rather than interpreted as a simple scalar measure of susceptibility.

2.4. *Scenario analysis and simulation evidence*

Agent-based models are especially useful when aggregate outcomes arise from heterogeneous local decisions and feedback (Macal and North, 2010; Railsback and Grimm, 2019). Their credibility depends on transparent specification, verification, validation, and disciplined experimentation. The ODD protocol encourages explicit descriptions of purpose, entities, state variables, process scheduling, and submodels (Grimm et al., 2020). Monte Carlo replication separates persistent tendencies from individual stochastic realisations, and common random numbers can improve contrasts when treatment and control respond similarly to shared randomness (Law, 2015).

Scenario analysis does not predict a single future. It asks conditional questions: if a source copied a specified set of attributes at a specified time, and if users retained evidence for a specified horizon, what would the implemented mechanisms produce? The strength of the answer depends on internal validity and sensitivity analysis. External claims require empirical calibration. In this project, a calibration interface exists, but observed repost, exposure, credibility, and uncertainty targets are not part of the repository. We therefore report mechanism-based simulation contrasts and state empirical calibration as future work.

3. The FAKENEWS-ABM model

3.1. *Purpose, entities, and environment*

FAKENEWS-ABM represents repeated source evaluation and reposting on a generic social-network site. Its purpose is to compare how source attributes, social recommendations, memory, reach, and timed interventions affect source selection and realised fake-news dissemination. The model is not tied to a specific platform or population and is not calibrated to reproduce an observed national prevalence. It is a mechanism-oriented computational laboratory.

The environment contains 400 SNS-user agents and four source types: `TRADITIONAL_MEDIA`, `UNKNOWN_MEDIA`, `FAKE_NEWS_SOURCE`, and `MIXED_SOURCE`. Each run lasts 400 periods. The source types differ in their attribute distributions, initial reach, and probability of publishing a false item. The configured false-publication probabilities are 0.092, 0.206, 0.667, and 0.416, respectively. At each period, every source independently produces a binary truth state. A user selection of a source is interpreted as reposting that period’s item from that source.

User agents have weights for source attributes and a WOM weight. Source attributes

Table 1. Core simulation configuration and interpretation.

Component	Value	Interpretation
SNS users	400	Decision-making agents in every run
News sources	4	Traditional, unknown, fake-news, and mixed source types
Periods	400	Sequential opportunities to evaluate and repost
Primary window	301–400	Late-run average used for selection and fake-repost shares
Memory	5, 10, 25, 50, 100, ∞	Number of periods for which endorsements remain active
Scenario timing	1, 25, 50, 100	First period in which copied attributes become active
WOM outcome	$-1, 0, +1$	Penalise, ignore, or reward a recommendation after truth is known

are represented by two-level probability distributions and cover positive and negative evoked emotion, simple language, political, geographic and social proximity, sensationalism, information quality, information entertainment value, source credibility, audiovisual material, hashtags, and links. These dimensions are paired by name with user weights. The setup allows the same observed source characteristic to be attractive to one user and less influential to another.

3.2. *Scheduling and source selection*

Each period follows a fixed schedule. First, sources produce their period-specific true or false states. Each user then evaluates known sources using active endorsements, applies the stochastic source-selection rule, and records one selected source when reach permits. The reporting subsystem stores selections per source, source truth states, and the cumulative number of distinct reposters. Timed scenarios update source attributes at their configured starting period. After decisions and reports, WOM recommendations are generated for use by contacts in the subsequent decision process.

The evaluation process converts source attribute distributions and user weights into endorsement events. Let $E_{u,s,t}$ denote the set of active endorsements that user u holds for source s at period t. A memory horizon m retains endorsement e when its period is newer than $t - m$; $m = -1$ denotes infinite memory. The source score is an aggregation of active evidence. Selection is stochastic rather than a deterministic maximum, allowing repeated runs to explore the distribution induced by attribute sampling and social feedback.

When source reach is disabled, every user selects exactly one source per period, so the four source counts sum to 400. When reach is enabled, a user may know no eligible source and contribute no selection. The denominator nevertheless remains the 400-agent population. This convention treats non-exposure as absence of reposting and prevents a reduced exposed population from mechanically inflating shares.

3.3. *WOM and outcome-specific policies*

WOM operates through contacts. A recommending user supplies evidence about a recently selected source. The receiver aggregates friend evaluations and deterministically identifies the strongest recommendation. If the source is previously unknown, it is added once to the receiver's known-source set. The model then applies the configured outcome policy after identifying whether the recommended item was false or true.

The policy values are `PENALIZE` (-1), `IGNORE` (0), and `REWARD` ($+1$). The signed policy is distinct from the user's WOM weight: the policy sets the direction of evidence, whereas the user attribute determines its importance in subsequent evaluation. An ignored outcome creates no endorsement. This implementation supports four central experimental regimes: false penalised or ignored, crossed with true rewarded or ignored. A positive treatment of false news is retained as a deliberately adverse counterfactual.

Three baseline WOM configurations isolate the mechanism. B0 disables WOM. B1 enables WOM but ignores both false and true outcomes, preserving discovery without outcome evidence. B2 enables WOM, penalises false outcomes, and rewards true ones. A B1–B0 contrast isolates discovery under the baseline source ecology, whereas B2–B1 isolates truth-sensitive outcome feedback.

3.4. *Source credibility and false-publication generation*

Credibility is both a perceived source attribute and an input to false-news generation in the current implementation. A source's probability of publishing fake news is obtained from its credibility distribution. This coupling is consequential. If a fake-news source copies a traditional source's credibility, its content-generation probability also becomes traditional-like. A larger selection share may then coexist with fewer false reposts.

The model records every source's binary false-news state in every period. This report was introduced specifically to prevent interpretation from relying on source identity alone. A selection from `FAKE_NEWS_SOURCE` is not automatically classified as a false repost; it is false only when that source's state in the same period equals one. Conversely, a selection from a traditional or mixed source contributes to the fake-repost count if that source published a false item in that period.

For run r, the target-source selection share over the final 100 periods is

$$S_r = \frac{1}{100} \sum_{t=301}^{400} \frac{N_{\text{target},t}}{400}, \tag{1}$$

where $N_{s,t}$ is the number of users selecting source s. The corresponding realised fake-repost share is

$$F_r = \frac{1}{100} \sum_{t=301}^{400} \frac{\sum_s N_{s,t} I_{s,t}^{\text{false}}}{400}, \tag{2}$$

where $I_{s,t}^{\text{false}}$ is the source's binary publication state. The cumulative count extends the numerator over all 400 periods without dividing by population or time.

3.5. *Scenario mechanism*

Scenarios copy attributes from a donor source A to a target source B at a configured period. The Excel scenario sheet names the donor, target, start period, and optional list of attributes. An explicit list copies only those dimensions. An empty list is a shortcut for copying all attributes. The experiment harness creates temporary workbooks and invokes the unchanged Java command-line entry point, keeping experiment-specific concerns outside the production simulation core.

We define three copy scopes. *Credibility-only* copies just source credibility and therefore primarily diagnoses the coupled false-publication mechanism. *Non-credibility* copies every source attribute except credibility; it changes engagement-related competitiveness while preserving the target's original false-publication propensity. *All-copy* copies every source attribute and represents complete imitation. The non-credibility treatment is the primary dissemination scenario. The other two are necessary to understand why target selection and false reposting can move in opposite directions.

3.6. *Implementation and reproducibility*

The reference implementation is written in Java. Configuration and source attributes are loaded from Excel workbooks, while the experiment harness uses Python and shell scripts to produce isolated workbook variants and execute the generic CLI. An experiment-only launcher seeds Java's pseudo-random source so that treatment and control can share initial random streams. The analyzer validates workbooks, derives run-level metrics, performs matched comparisons, and saves CSV summaries and static figures. The repository is publicly available at https://github.com/pleger/FAKENEWS-ABM. The exact experiment directories and commands are stated in Section 4 and the data-availability declaration.

4. Experimental design and analysis

4.1. *Phased design*

The analysis consolidates three completed experiment roots while preserving their different purposes. The first root, `research_questions_20260812_180619`, contains 814 run-level records from exploratory questions on copy scope, memory and timing, WOM under network variation, and seeded replication. The second, `dissemination_experiments_20260813_215632`, contains 178 baseline runs. The third, `dissemination_experiments_20260814_083018`, contains 1,911 strategy, confirmation, and robustness runs. Together they provide 2,903 simulation runs.

The phases are used hierarchically. Baseline checks test whether the implementation produces expected publication rates and a directional WOM response. The broad strategy grid maps interactions and ranks candidate conditions. The confirmation phase repeats six policy-specific non-credibility contrasts with 30 common seeds. Robustness analyses vary supported network opportunity and reach. The exploratory phase supplies additional memory values, WOM-on/off contrasts, and evidence about seed pairing. We do not pool all runs into a single regression because configurations and estimands differ across phases.

Table 2. Experiment phases consolidated in this study.

Phase	Purpose	Runs
Exploratory questions	Copy scope, memory/timing, WOM/network, and common-seed precision	814
Baseline validation	Publication-rate checks, no WOM, discovery-only WOM, and truth-sensitive WOM	178
Strategy matrix	Three copy scopes, three memories, three false-news policies, and four timings	1,287
Confirmation	Six false/true WOM outcome policies with 30 common seeds	360
Robustness	Three contact levels and four reach conditions	264
Total	Distinct simulation runs analysed	2,903

4.2. *Baseline design*

The baseline uses 30 common seeds, numbered 1001–1030, for three conditions at memory 25. B0 disables WOM and uses the default signed outcome policy, which is inactive because no recommendations occur. B1 enables WOM but sets false and true outcomes to zero. B2 enables WOM, penalises false outcomes (-1), and rewards true outcomes ($+1$). All three use 15 contacts, a friends proportion of 0.33, and disabled source reach. The baseline also contains memory-sensitivity pairs at memory 10, 50, 100, and infinite with 11 seeds per condition, contrasting discovery-only and truth-sensitive WOM.

Five a priori checks define baseline success. Four compare observed source-specific false-publication rates with configured values using an absolute tolerance of 0.02. The fifth requires the upper bound of the paired 95% confidence interval for B2 minus B1 fake-repost share to be below zero. B1 minus B0 is reported but was not required to be non-zero.

4.3. *Strategy matrix*

The strategy phase uses memories 25, 100, and infinite; false-news WOM outcomes -1, 0, and $+1$; true-news outcome $+1$; and activation periods 1, 25, 50, and 100. For every memory-policy combination, 11 common seeds are run for a no-scenario control and for credibility-only, non-credibility, and all-copy treatments. Each treatment is compared only with the control that has the same memory, WOM policies, contact configuration, reach configuration, and seed. This produces 1,287 runs and 108 paired treatment effects.

The confirmation phase fixes memory at 25, activates non-credibility copying in period 1, and crosses false-news outcomes -1, 0, and $+1$ with true-news outcomes 0 and $+1$. Each of the six treatment-control pairs uses seeds 1001–1030, giving 360 runs. This design tests whether the large differences associated with ignoring true recommendations persist with more replications.

4.4. *Robustness and exploratory matrices*

The robustness phase fixes non-credibility copying at period 1, memory 25, false-news penalisation, and true-news reward. It varies the number of contacts across 6, 15, and 27. For each contact count, source reach is either disabled or enabled with target reach 14.7%, 46.7%, or 81.2%. Each treatment and matched control uses 11 common seeds, producing 264 runs and 12 paired effects. The model has no homophily parameter, so the experiment does not claim a homophily sensitivity result.

The earlier exploratory matrix examines engagement-only and all-copy scenarios under memory 25 and infinite, with WOM on or off. Its memory/timing grid contains memories 5, 10, 25, 50, 100, and infinite crossed with activation periods 1, 25, 50, and 100. Its network matrix compares WOM on and off for realised contact degrees 1, 4, and 8, with source reach disabled or enabled. Finally, 20 conditions are rerun with common seeds to compare rank ordering and paired precision against earlier unpaired estimates.

4.5. *Metrics and estimands*

The primary metric is realised fake-repost share in periods 301–400, F_r. Secondary metrics are target-source selection share in the same window, cumulative fake reposts across all periods, target-source false-publication rate, and the number of unique target-source reposters at period 400. The final-100 window was chosen before the consolidated analysis to emphasise sustained late-run behaviour while retaining the cumulative metric for total burden.

For each treatment T and matched control C, run-level differences are constructed by seed:

$$D_i = Y_{T,i} - Y_{C,i}, \qquad i = 1, \ldots, n. \tag{3}$$

The reported effect is $\bar{D}$. Its standard error is $s_D/\sqrt{n}$, and the 95% interval is

$$\bar{D} \pm t_{0.975,n-1} \frac{s_D}{\sqrt{n}}. \tag{4}$$

The archived analyzer initially displayed normal-approximation intervals using 1.96. We independently recomputed headline intervals with Student's t distribution for this manuscript; point estimates are unchanged. Because the strategy grid contains many comparisons, its ranking is descriptive and exploratory. No multiplicity correction is applied. Confirmatory emphasis is placed on the six pre-specified policy cells with 30 pairs.

4.6. *Validation and quality controls*

Every new-suite output workbook was required to contain exactly 400 period rows in the repost, fake-news-state, and unique-reposter sheets. Simulation-period keys had to be unique, source truth states binary, and selection totals within 0–400. With source reach disabled, the source totals had to equal 400 in every period. Metadata in the manifest was matched to each workbook, and expected run counts were checked: 178 baseline, 1,287 strategy, 360 confirmation, and 264 robustness runs.

Table 3. Baseline validation and paired WOM contrast. Publication rates are averages across truth-sensitive baseline runs.

Quantity	Expected	Observed	Status
Traditional-source false-publication rate	9.20%	9.26%	Pass
Unknown-source false-publication rate	20.60%	20.58%	Pass
Fake-news-source false-publication rate	66.70%	67.24%	Pass
Mixed-source false-publication rate	41.60%	41.64%	Pass
Truth-sensitive minus discovery-only fake-repost share	< 0	-5.26 pp	Pass

The baseline root passed all five behavioural checks. The strategy root does not repeat baseline conditions and therefore reports zero baseline checks by design; its 1,911 workbooks all passed structural loading and metric construction. The exploratory root separately reports 274 workbooks and 814 internal simulation runs with balanced condition coverage. The distinction occurs because earlier workbooks contain multiple repetitions, whereas the seeded launcher writes one run per workbook.

Common random numbers improve precision only if shared randomness induces positive correlation between treatment and control. Scenario branches can cause random streams to diverge later, so pairing is not assumed to help universally. We therefore report the ratio of paired to prior unpaired standard errors. A ratio below one indicates improvement. Rank reproduction is assessed with Spearman's ρ across the 20 shared conditions.

All analysis is descriptive of the implemented model. We use causal language only for within-model interventions whose treatment-control difference is mechanically defined by the scenario. We do not interpret the Monte Carlo intervals as uncertainty about a real population, and we do not claim empirical calibration.

5. Results

5.1. *Baseline validation and truth-sensitive WOM*

All five baseline checks passed. The observed false-publication rates were 9.26% for traditional media, 20.58% for unknown media, 67.24% for the fake-news source, and 41.64% for the mixed source. Each value was within 0.02 of its configured probability (Table 3). These checks establish that the saved period-level truth states reproduce the intended source ecology closely enough for the subsequent repost metric.

Figure 1 compares late-run fake-repost shares. B0 (no WOM) and B1 (discovery-only WOM) both averaged 15.36%. Their paired difference was exactly zero for the retained metrics. Under this configuration, allowing contacts to reveal sources without creating outcome endorsements did not alter aggregate selection. B2 (false penalised, true rewarded) averaged 10.11%. The B2–B1 paired effect was -5.26 percentage points (Student 95% CI $[-5.95, -4.56]$, $n = 30$). Target-source selection changed by less than 0.001 percentage points and its interval included zero. Thus truth-sensitive WOM re-

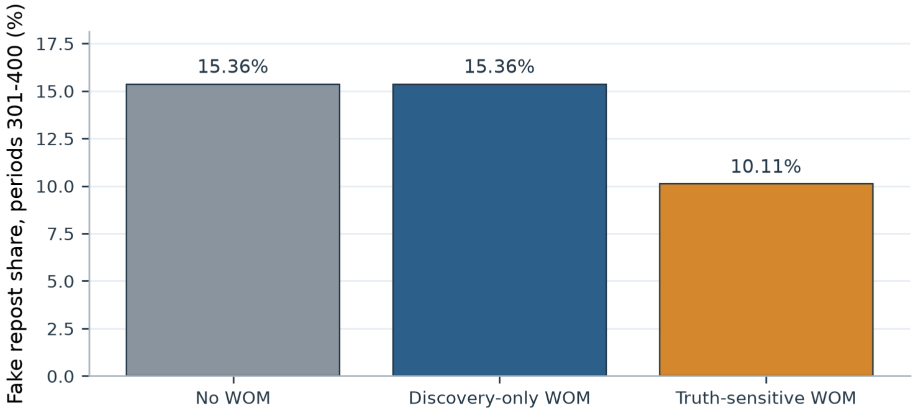

Figure 1. Baseline WOM comparison. Bars show mean realised fake-repost share in periods 301–400 across 30 common seeds. Discovery-only WOM ignores both truth outcomes; truth-sensitive WOM penalises false and rewards true recommendations.

duced system-level false reposting primarily through source-choice composition rather than a detectable late-run change in the already small target-source share.

Memory sensitivity shows that the protective magnitude is configuration dependent. The truth-sensitive minus discovery-only difference is particularly large at memory 10 and remains negative at memory 50, 100, and infinite. These comparisons support the direction of the baseline mechanism but also warn against treating 5.26 points as a universal constant.

5.2. *Copy scope separates popularity from false dissemination*

The exploratory copy-scope experiment provides the clearest descriptive example. With infinite memory and WOM active, the no-scenario target received 0.81% of selections and published false items in 67.36% of the final 100 periods. Engagement-only copying increased selection to 4.29% while the target false-publication rate remained 65.82%. Full copying increased selection further, to 9.22%, but lowered the target false-publication rate to 9.82%. Full imitation therefore made the target more popular and simultaneously transformed its content process into one resembling the traditional donor.

Figure 2 places target selection and target false-publication rate on separate axes. The engagement-only point moves mostly to the right, whereas all-copy moves right and sharply downward. This pattern motivates non-credibility copying as the primary camouflage treatment in the later suite.

The broader strategy matrix confirms the distinction (Figure 3). The largest target-selection increase was produced by all-copy at period 1, infinite memory, false-news penalisation, and true-news reward: +8.54 points (95% CI $[7.83, 9.25]$, $n = 11$). Yet its actual fake-repost effect was -1.16 points (95% CI $[-1.37, -0.95]$), and cumulative false reposts decreased by 2,572. All-copy points therefore occupy the high-selection, lower-false-repost region.

By contrast, non-credibility points are predominantly in the quadrant where both target selection and fake reposting increase. The strongest 11-seed strategy cell used period 1, infinite memory, false-news penalisation, and true-news reward. It increased target selection by 3.50 points (95% CI $[3.24, 3.76]$), fake reposts by 1.76 points (95% CI

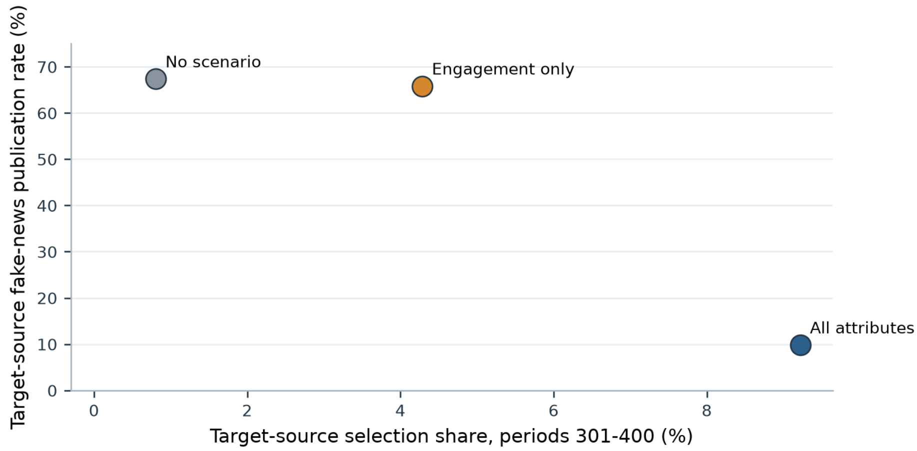

Figure 2. Copy-scope comparison under infinite memory and active WOM in the exploratory phase. Target selection and target false-publication rate are distinct outcomes.

Table 4. Selected paired scenario effects relative to matched controls. Confidence intervals use Student's t distribution.

Condition	Fake reposts	Target selection	Cumulative false reposts
Non-credibility, P1, M25, false -1/true 0	+2.89 pp	+5.64 pp	$+4,354$
Non-credibility, P1, M∞, false -1/true $+1$	+1.76 pp	+3.50 pp	$+2,737$
Non-credibility, P1, M100, false -1/true $+1$	+0.93 pp	+1.88 pp	$+2,239$
All-copy, P1, M∞, false -1/true $+1$	-1.16 pp	+8.54 pp	$-2,572$

$[1.58, 1.93]$), and cumulative false reposts by 2,737. Copying credibility alone generally reduced false reposting, again reflecting the content-generation coupling.

5.3. *Memory and intervention timing*

The exploratory memory/timing grid shows a strong interaction (Figure 4). With memory 5, target-selection effects were essentially zero at all activation times. At memory 10, activation in period 1 added 0.16 points, while later effects were weak. From memory 25 upward, period 1 was consistently strongest and effects declined at periods 25, 50, and 100. The maximum exploratory target-selection effect was 8.34 points for memory 100 and period 1; infinite memory at period 1 produced 8.20 points.

The new strategy matrix yields the same qualitative timing result for the primary non-credibility outcome. With infinite memory and the $-1/+1$ policy, the fake-repost effect falls from 1.76 points at period 1 to 0.66 at period 25, 0.43 at period 50, and 0.26 at period 100. At memory 100, the period-1 effect is 0.93 points. Memory 25 under the standard $-1/+1$ policy has a small final-window effect, a result that later motivated the 30-seed policy confirmation. Timing should therefore be interpreted jointly with the outcome policy: the same nominal memory does not generate the same treatment-control contrast when true outcomes are ignored rather than rewarded.

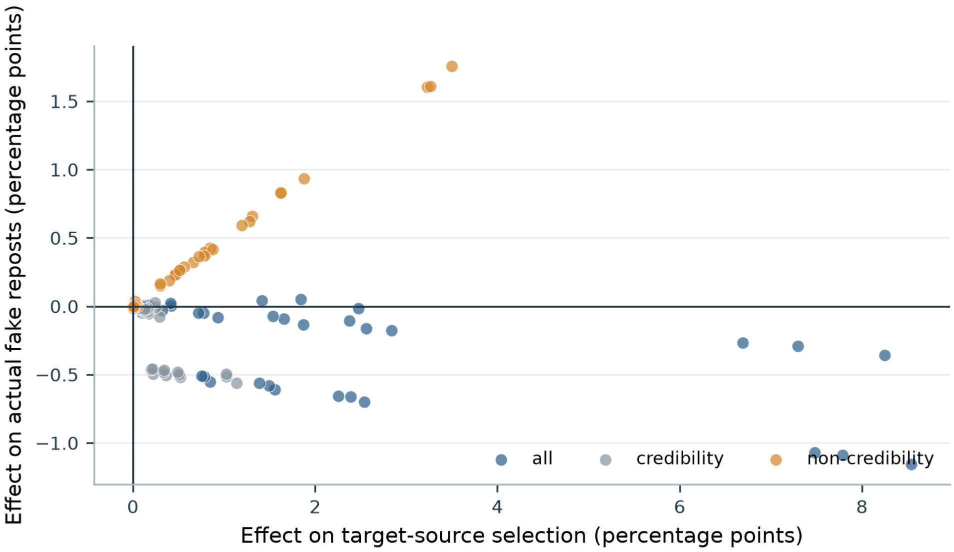

Figure 3. Paired strategy effects on target selection and actual fake reposting. Each point is a treatment relative to its matched control. The horizontal and vertical zero lines separate increased and decreased outcomes.

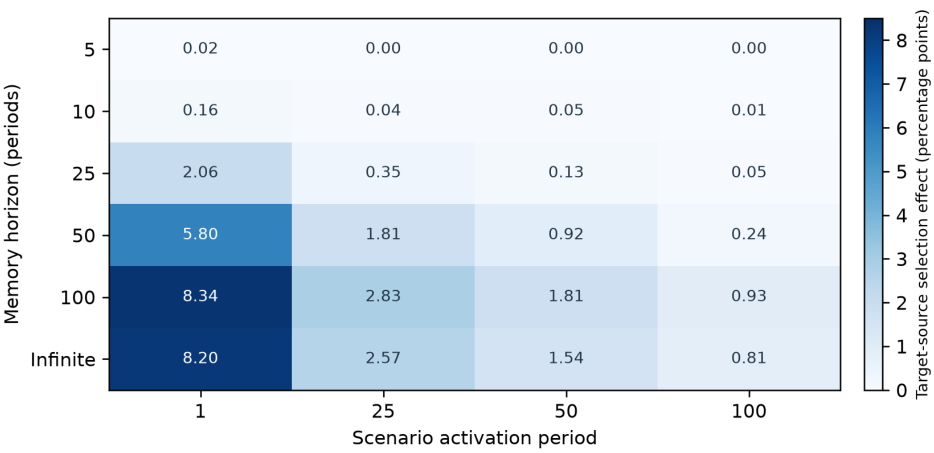

Figure 4. Exploratory interaction between memory horizon and scenario activation. Cell values are target-source selection effects in percentage points relative to a no-scenario control with the same memory.

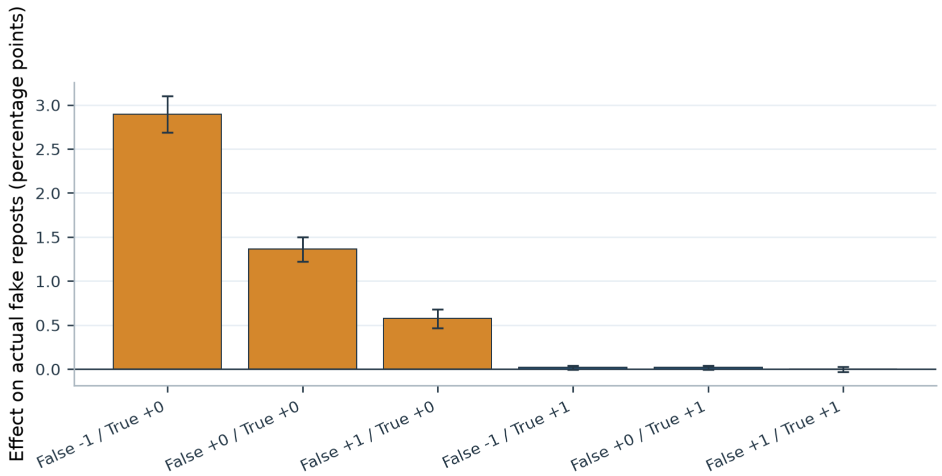

Figure 5. Confirmatory non-credibility camouflage effects under six WOM outcome policies. Bars show treatment-control effects on fake-repost share; error bars are paired Student 95% confidence intervals across 30 seeds.

5.4. *Outcome-specific WOM policies*

Figure 5 presents the six confirmatory policy contrasts for non-credibility copying at period 1 and memory 25. When true recommendations were ignored, the false-news rule produced a graded response. Penalising false recommendations yielded the largest treatment effect: +2.89 fake-repost points (95% CI $[2.69, 3.10]$), +5.64 target-selection points (95% CI $[5.27, 6.01]$), and 4,354 additional cumulative false reposts. Ignoring false outcomes yielded +1.36 fake-repost points, and rewarding false outcomes yielded +0.57 points. All three intervals were above zero.

When true recommendations were rewarded, the final-window treatment effects were close to zero. The $-1/+1$ condition produced +0.017 points (95% CI $[-0.008, 0.042]$), the $0/+1$ condition +0.017 points (95% CI $[-0.007, 0.041]$), and the $+1/+1$ condition -0.002 points (95% CI $[-0.032, 0.029]$). None of these intervals excluded zero. Cumulative effects remained positive, between approximately 766 and 859 false reposts, indicating earlier-period differences that were not sustained strongly in periods 301–400.

The apparently counterintuitive $-1/0$ ranking does not imply that penalising false news increases misinformation in absolute terms. Each point compares camouflage and control under the same policy. The baseline establishes that $-1/+1$ reduces false reposting relative to discovery-only WOM. The confirmation asks a different question: how much additional effect camouflage produces after the policy is fixed. Rewarding true outcomes appears to stabilise treatment and control sufficiently that their late-run difference is small, whereas ignoring true outcomes leaves more room for copied non-credibility attributes to separate them.

5.5. *Network and reach robustness*

The exploratory WOM-on/off comparison found negative target-selection effects in all six supported degree/reach cells. WOM reduced transformed-source share by 1.42–2.67 points and reduced fake-repost share by 1.61–6.19 points. These intervals were below zero for target selection. Under that estimand, WOM was unfavourable to the

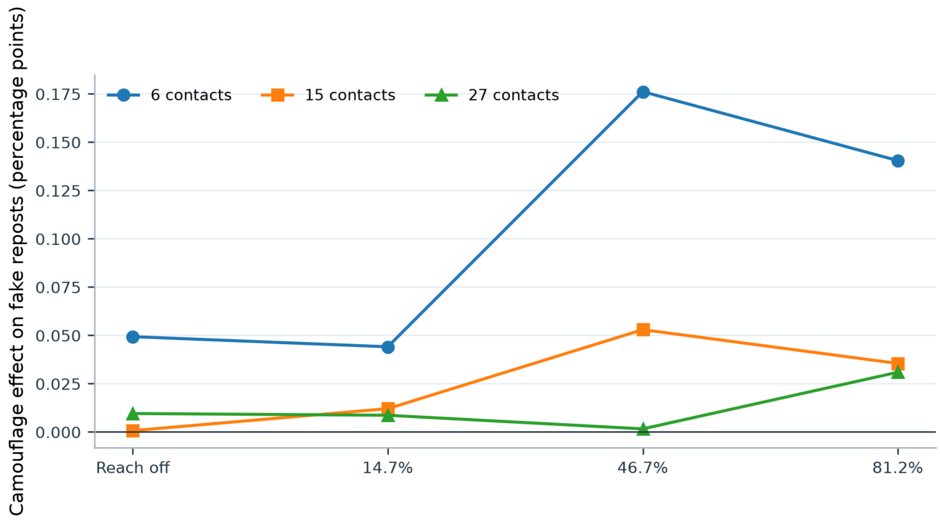

Figure 6. Robustness of non-credibility camouflage effects across contact and target-reach settings. Values are paired effects on actual fake-repost share.

transformed source and favourable to the system-level misinformation outcome.

The later robustness phase estimates a different quantity: non-credibility camouflage minus a matched no-scenario control when truth-sensitive WOM is already enabled. Its fake-repost effects range from approximately 0.00 to 0.18 points (Figure 6). The largest effect occurs with six contacts and 46.7% target reach: +0.176 points (95% CI $[0.082, 0.270]$). Several cells have intervals crossing zero. Cumulative effects are positive in all 12 cells, ranging from 301 to 2,334 additional false reposts, because small period-level differences accumulate over 400 periods.

These results are not contradictory. WOM-on/off measures the total contribution of social recommendations, while camouflage/control measures the incremental contribution of copied attributes inside an active WOM regime. Table 5 summarises the distinct contrasts.

5.6. *Seeded reproduction and precision*

Common-seed runs reproduced the earlier condition ordering. Across 20 memory, WOM, and timing cells, Spearman's ρ was 0.998 for target-source share and 0.974 for fake-repost share. Both associations were statistically distinguishable from zero. Pairing reduced the estimated WOM-effect standard error in eight of ten cells (Figure 7); the median paired-to-unpaired standard-error ratio was 0.789. The two exceptions were memory 25 at period 1 (ratio 1.22) and memory 25 with no scenario (ratio 2.03), where stream divergence and a near-zero effect limited the benefit of pairing.

6. Discussion

The central result is not a single optimal parameter combination but a measurement distinction. Source popularity and misinformation dissemination can move in opposite directions. Full copying produced the largest increase in target-source selection, yet it reduced realised fake reposting because it transferred credibility and thereby lowered

Table 5. Range and interpretation of robustness results.

Contrast	Observed range	Interpretation
Exploratory WOM on minus WOM off	−2.67 to −1.42 pp target selection	WOM was unfavourable to the transformed source in all six supported network cells
Camouflage minus matched control	0.00 to +0.18 pp fake reposts	Camouflage effects persisted weakly under some reach/contact configurations
Common-seed rank reproduction	ρ = 0.998 selection; ρ = 0.974 fake reposts	Condition ordering was highly reproducible
Paired precision	Lower SE in 8 of 10 cells	Common seeds usually narrowed Monte Carlo uncertainty

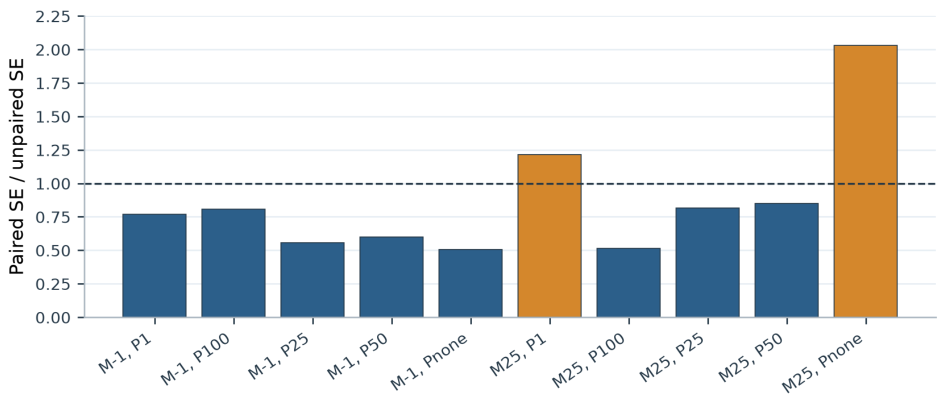

Figure 7. Ratio of paired to prior unpaired standard errors. Values below one indicate a precision gain from common seeds.

the target's probability of producing false content. Non-credibility copying produced a smaller popularity gain but a larger misinformation increase because it preserved the source's editorial propensity. Studies that label every selection of a "fake-news source" as a fake repost would miss this difference and overstate the harm of full copying while potentially understating camouflage.

This finding has a direct modelling implication. Perceived credibility and objective false-publication probability should ideally be distinct state variables. Their coupling in the current implementation made the diagnostic possible, but it also caused one scenario parameter to change both user perception and source behaviour. Separating them would support interventions such as improving reputation without changing editorial conduct, or changing editorial conduct without immediate reputation repair. Until that refactoring is made, non-credibility copying is the cleanest available dissemination treatment and credibility/all-copy scenarios should be labelled mechanistic diagnostics.

The baseline supports the internal logic of truth-sensitive WOM. Discovery alone did not alter the retained outcomes, whereas penalising false and rewarding true recommendations reduced fake reposting substantially. This is consistent with research showing that attention to accuracy and source quality can improve information choice (Pennycook and Rand, 2019; Pennycook et al., 2020; Epstein et al., 2021). The comparison remains model specific: it does not estimate the effect of a platform fact-check or a real interpersonal correction. It shows that signed truth feedback, when integrated into an endorsement-based decision process, can shift aggregate repost composition.

The confirmation results expose why policy labels must be interpreted relationally. The largest camouflage-control difference occurred under false penalisation and no true reward. At first glance this seems to conflict with the protective baseline. It does not. The baseline compares policies, while confirmation compares scenarios within a policy. Under $-1/+1$, both treatment and control benefit from positive evidence attached to true outcomes, and their late-run difference is small. Under $-1/0$, that stabilising pathway is absent; copied engagement attributes create a much larger relative separation even though false outcomes are penalised in both branches. Future work should decompose the endorsement stock over time to determine precisely how positive true-news WOM, negative false-news WOM, and non-WOM source attributes contribute to the score trajectories.

Memory and timing behave as coupled path-dependence mechanisms. A very short memory removes scenario-generated evidence before it can accumulate, while an early intervention under long memory shapes a large fraction of the run. Delayed interventions have less time to alter the evidence distribution and must compete with pre-existing endorsements. These patterns resemble cumulative-advantage processes in which early social signals can produce persistent inequality (Salganik et al., 2006; Muchnik et al., 2013). However, the model does not imply that longer human memory necessarily increases vulnerability. Infinite model memory preserves every form of evidence, including negative evidence and control history. Its effect depends on what evidence the scenario and WOM policy generate.

The two network analyses illustrate a broader lesson about robustness. An effect is meaningful only relative to its control. WOM-on/off contrasts show that truth-sensitive social exchange can suppress the transformed source across supported contact and reach cells. Camouflage/control contrasts show that copied attributes can still add a small burden after WOM is enabled. Reporting either result without its estimand would create an apparent contradiction. For platform design, the distinction corresponds to asking whether social feedback is beneficial overall versus whether a manipulation remains effective after social feedback exists.

The common-seed analysis improves confidence in the ordering of conditions. Near-perfect rank reproduction indicates that the major parameter patterns are not artefacts of a particular independent Monte Carlo sample. Pairing usually narrows uncertainty, but not automatically. Treatment branches change later random calls, and nearly zero effects can have unstable standard-error ratios. The experiment harness therefore records seeds and reports empirical precision gains rather than assuming them.

6.1. *Implications for research and practice*

For misinformation research, the study supports a three-part outcome framework: audience share, editorial false-publication propensity, and realised false reposts. These quantities answer different questions and should be reported together. For journalism, a source that adopts familiar professional conventions may gain legitimacy without changing its truthfulness. For communication and marketing research, engagement attributes can transfer attention, but the effect depends on when they are introduced and on the audience's retained evidence. For platform engineering, recommendation exposure, outcome evaluation, and source reach should be logged as separate mechanisms.

The model may also support countermeasure experiments. Media-literacy interventions can be represented by changing user weights on credibility, information quality, sensationalism, and emotion. Platform friction can be represented by reduced reach or delayed transmission. Reputation systems can be represented by truth-sensitive WOM with different magnitudes. Yet such scenarios should be tested only after separating perceived credibility from objective false-publication probability, because otherwise an intervention that changes perceived trust may accidentally rewrite source behaviour.

6.2. *Limitations*

First, the model is not empirically calibrated. The parameter values originate from the project's input workbook and earlier modelling design, not a joint fit to platform exposure and repost data. Monte Carlo intervals describe stochastic variation conditional on those values; they are not population confidence intervals. The repository includes a calibration template and weighted-distance tool, but no empirical targets, sampling uncertainties, or held-out observations.

Second, the network representation is limited. Contact opportunity and a binary source-reach mechanism are supported, but homophily is absent. Echo-chamber dynamics documented in empirical research (Del Vicario et al., 2016) cannot be evaluated by the current CLI harness. Degree is also represented through a compact contact mechanism rather than a validated platform network topology.

Third, sources are types rather than evolving organisations. They do not strategically choose truth states, react to moderation, or adapt attributes in response to audience feedback. The scenario is exogenous and deterministic once its activation period is reached. Users likewise do not create original content, verify claims independently, or leave the platform.

Fourth, the factorial strategy ranking involves 108 effects and no correction for multiple comparisons. We therefore treat rankings as exploratory and emphasise the 30-seed confirmation for policy claims. Even there, the six conditions were selected after inspecting earlier patterns, so a future preregistered replication would provide stronger confirmatory evidence.

Finally, common random numbers do not guarantee paired trajectories. Once a treatment changes source selection, subsequent random-number consumption can diverge. Pairing remains useful empirically, but the observed precision improvement should not be interpreted as perfect counterfactual synchronisation.

6.3. *Next research steps*

The next technical step is to separate perceived credibility, reputation, and objective false-publication probability. The next empirical step is to obtain targets for source exposure, repost rates conditional on exposure, source-level truthfulness, and credibility judgments. Calibration should estimate WOM weight, memory horizon, reach, and the endorsement aggregation scale using training targets, then evaluate predictive ordering on held-out targets. Finally, a homophily mechanism and alternative network topologies should be introduced through an explicit model extension rather than simulated indirectly by workbook manipulation.

7. Conclusions

This study used 2,903 agent-based simulation runs to examine how source imitation, memory, timing, and WOM shape fake-news dissemination. The validated baseline reproduced configured false-publication rates and showed that truth-sensitive WOM reduces actual fake reposting relative to discovery-only WOM. The scenario analysis then demonstrated that source popularity is not a sufficient misinformation metric. Full attribute copying maximised the target source's selection gain but reduced false reposts because credibility also changed the target's publication process. Non-credibility copying preserved the target's high false-news propensity and provided the more defensible camouflage treatment.

The most concerning confirmed combination acted in period 1, retained endorsements for 25 periods, copied every non-credibility attribute, penalised false recommendations, and ignored true recommendations. It increased final-window fake reposts by 2.89 percentage points relative to its matched control. The result is relational rather than a claim that false-news penalties are harmful: the baseline shows the protective role of a balanced $-1/+1$ truth-sensitive policy. Early interventions and longer memory generally strengthened scenario effects, while very short memory largely erased them.

The experiments also show the value of explicit controls and reproducible randomness. Network/WOM and camouflage/control contrasts answer different questions; common seeds reproduced condition rankings and usually improved precision. These practices help prevent attractive but misleading conclusions from a stochastic scenario grid.

FAKENEWS-ABM should be understood as a conditional theory made executable. Its present evidence identifies plausible mechanisms, clarifies measurement, and supports future hypotheses. Empirical calibration, a separate objective truth-generation parameter, homophily, and preregistered replication are required before using it for platform-specific prediction. Within those boundaries, the model shows that the riskiest imitation is not necessarily copying everything. It is copying the attributes that attract attention while preserving the underlying propensity to publish false news.

Acknowledgements

This work was developed in the context of the PLURALISMO project PLU230018, *Evaluación de Estrategias para Diseminar Fake News Usando Inteligencia Artificial*. The authors thank the contributors to the earlier endorsement-based simulation studies on which the reasoning architecture builds.

Disclosure statement

The authors report no competing interests. This statement must be confirmed by all authors before submission.

Funding

Project identifier PLU230018 is documented in the software repository. The formal funder name, award wording, and any required grant acknowledgement must be confirmed before journal submission.

Data and code availability

The FAKENEWS-ABM source code and experiment harness are available at `https://github.com/pleger/FAKENEWS-ABM`. The analyses reported here use these experiment roots:

- `research_questions_20260812_180619`;
- `dissemination_experiments_20260813_215632`; and
- `dissemination_experiments_20260814_083018`.

Before submission, the exact code commit and generated analysis tables should be archived in a versioned release or research repository with a persistent identifier.

Author contributions

The final CRediT contribution statement requires approval from all authors. Proposed roles to verify are: conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing—original draft, writing—review and editing, visualisation, supervision, project administration, and funding acquisition. No individual allocation is asserted in this draft without author confirmation.

Ethics statement

The study analyses synthetic agents and simulation outputs and does not use human participants or personal data. Any future empirical calibration dataset must undergo its own ethical and privacy review.

References

Eytan Bakshy, Solomon Messing, and Lada A. Adamic. Exposure to ideologically diverse news and opinion on Facebook. *Science*, 348(6239):1130–1132, 2015. .

Damon Centola. The spread of behavior in an online social network experiment. *Science*, 329(5996):1194–1197, 2010. .

Paul R. Cohen. *Heuristic Reasoning about Uncertainty: An Artificial Intelligence Approach*. Pitman, Boston, 1985.

Michela Del Vicario, Alessandro Bessi, Fabiana Zollo, Fabio Petroni, Antonio Scala, Guido Caldarelli, H. Eugene Stanley, and Walter Quattrociocchi. The spreading of misinformation online. *Proceedings of the National Academy of Sciences*, 113(3): 554–559, 2016. .

Ziv Epstein, Adam J. Berinsky, Rocky Cole, Andrew Gully, Gordon Pennycook, and David G. Rand. Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3), 2021. .

Volker Grimm, Steven F. Railsback, Christian E. Vincenot, Uta Berger, Cara Gallagher, Donald L. DeAngelis, Bruce Edmonds, Jiaqi Ge, Jarl Giske, Jürgen Groeneveld, Alice S. A. Johnston, Alexander Milles, Jacob Nabe-Nielsen, J. Gary Polhill, Viktoriia Radchuk, Marie-Sophie Rohwäder, Richard A. Stillman, Jan C. Thiele, and Daniel Ayllón. The ODD protocol for describing agent-based and other simulation models: a second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, 23(2):7, 2020. .

Nir Grinberg, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer. Fake news on Twitter during the 2016 U.S. presidential election. *Science*, 363 (6425):374–378, 2019. .

Andrew Guess, Jonathan Nagler, and Joshua Tucker. Less than you think: Prevalence and predictors of fake news dissemination on Facebook. *Science Advances*, 5(1): eaau4586, 2019. .

Averill M. Law. *Simulation Modeling and Analysis*. McGraw-Hill Education, 5 edition, 2015.

David M. J. Lazer, Matthew A. Baum, Yochai Benkler, Adam J. Berinsky, Kelly M. Greenhill, Filippo Menczer, Miriam J. Metzger, Brendan Nyhan, Gordon Pennycook, David Rothschild, Michael Schudson, Steven A. Sloman, Cass R. Sunstein, Emily A. Thorson, Duncan J. Watts, and Jonathan L. Zittrain. The science of fake news. *Science*, 359(6380):1094–1096, 2018. .

Charles M. Macal and Michael J. North. Tutorial on agent-based modelling and simulation. *Journal of Simulation*, 4(3):151–162, 2010. .

Lev Muchnik, Sinan Aral, and Sean J. Taylor. Social influence bias: A randomized experiment. *Science*, 341(6146):647–651, 2013. .

Gordon Pennycook and David G. Rand. Fighting misinformation on social media using crowdsourced judgments of news source quality. *Proceedings of the National Academy of Sciences*, 116(7):2521–2526, 2019. .

Gordon Pennycook, Jonathon McPhetres, Yunhao Zhang, Jackson G. Lu, and David G. Rand. Fighting COVID-19 misinformation on social media: Experimental evidence for a scalable accuracy-nudge intervention. *Psychological Science*, 31(7):770–780, 2020. .

Steven F. Railsback and Volker Grimm. *Agent-Based and Individual-Based Modeling: A Practical Introduction*. Princeton University Press, 2 edition, 2019.

Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts. Experimental study of inequality and unpredictability in an artificial cultural market. *Science*, 311 (5762):854–856, 2006. .

Oswaldo Terán, Paul Leger, and Manuela López. Modeling and simulating chinese cross-border e-commerce: an agent-based simulation approach. *Journal of Simulation*, 2022. .

Oswaldo Terán, Paul Leger, and Manuela López. Factors that drive market share and the oligopolistic character of cross-border B2C e-commerce: an agent-based scenario-analysis approach. *Simulation: Transactions of the Society for Modeling and Simulation International*, 2024. .

Soroush Vosoughi, Deb Roy, and Sinan Aral. The spread of true and false news online. *Science*, 359(6380):1146–1151, 2018. .